2.6.6 Wrap-Up: Cool Down

1. Use algebra tiles to find the product of $(x-5)^2$.

$(x-5)^2 =$ ____________________

Unit 2, Lesson 7: Multiplying Binomials and Polynomials

Benchmark

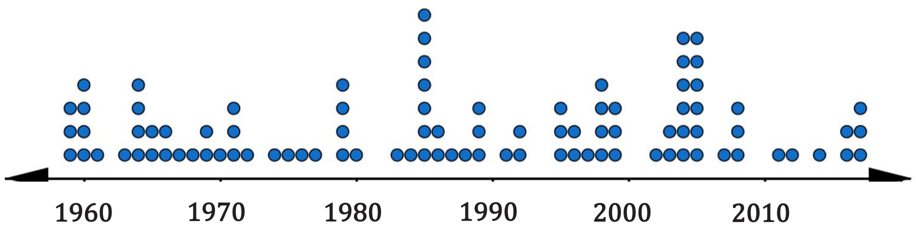
MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can multiply binomials.
- ☐ I can multiply polynomials.

2.7.1 Warm-Up: Notice and Wonder

1. The dot plot shows the number of hurricanes that made landfall in the continental United States from 1959 – 2017.



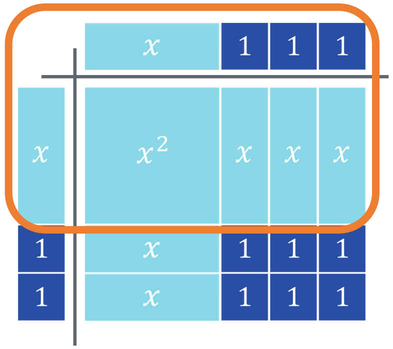
Number of Hurricanes 1959 - 2017

   a. What do you notice when looking at the dot plot?

   b. What do you wonder about the data presented?

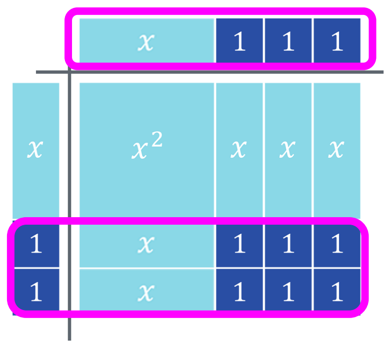
2.7.2 Guided Instruction: Multiplying Binomials

An algebra tiles model and an area model of $(x+2)(x+3)$ are shown. The orange box in both models shows that the x from the first factor was distributed to the $(x+3)$.



	x	$+3$
x	x^2	$3x$
$+2$		

The pink boxes show the 2 from $(x+2)$ being distributed to the $(x+3)$.



	x	$+3$
x	x^2	$3x$
$+2$	$2x$	6

Distributing the x and the 2 from $(x+2)$ to $(x+3)$ can be algebraically represented as $x(x+3) + 2(x+3)$.

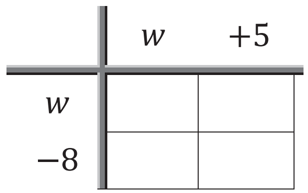
1. Apply the distributive property. Then combine like terms.

$$x(x+3) + 2(x+3)$$

2. Discuss with your partner how the algebra tile model or area model relates to applying the distributive property.

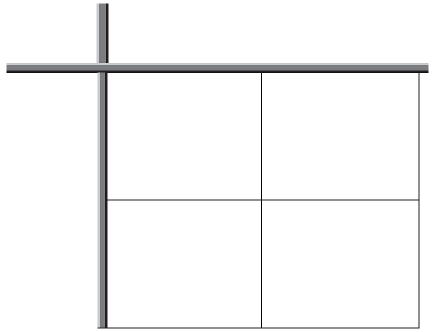
Brayden and Jackson were asked to find the product of $(w-8)(w+5)$ using different methods.

3. Complete the work for each student using the method they were assigned.

Brayden's Distributive Method	**Jackson's Area Model**
$w(w+5) - 8(w+5)$	

4. Multiply $(m^3 + 2k)(5m - 6k)$ using the distributive property.

5. Find the product $(2a + 3b)^2$ using an area model.



2.7.3 Guided Practice: Multiplying Binomials

Find each product using your preferred method. Show your work.

1. $(17 - x)(1.2x^2 + 6)$

2. $\left(\frac{1}{4}m + 13n\right)^2$

2.7.4 Collaboration: Multiplying Polynomials

The same methods can be extended to multiply any polynomial.

1. Work with your partner to complete the following.

   a. Multiply $(y - 1)(y^2 + 7x - 26)$ using the area model.

	y^2	$+7x$	-26
y			
-1			

$(y - 1)(y^2 + 7x - 26) =$ ________________________

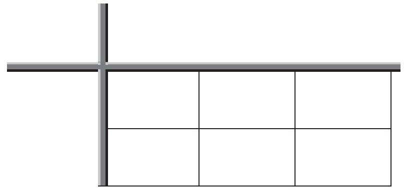
   b. Multiply $(1.5x - 5)(4x^2 - 7x + 3)$ using the distributive property.

$1.5x(4x^2 - 7x + 3) - 5(4x^2 - 7x + 3)$

   c. Compare your answers with your partner and resolve any differences.

2. Decide who will be Partner A and who will be Partner B.

   a. Find the product of $(2n^5 - n - 14)(3n + 2)$ using the method indicated.

Partner A Distributive Property	Partner B Area Model
$(2n^5 - n - 14)(3n + 2)$	

   b. Compare your answers with your partner and resolve any differences.

   c. Discuss any connections you notice between the methods used to find the product.

   d. Complete the statement.

   When multiplying polynomials, I prefer to use the

   - ○ area model
   - ○ distributive property

   because ...

2.7.5 Your Turn!

1. Find the product of each expression using your preferred method.

$\left(2x - \frac{1}{3}\right)\left(x + \frac{5}{4}\right)$	$\left(\frac{2}{3} + a\right)(-5a + 2a^2 - 8)$
$(7r - 0.18)(-6 + r^2 + 4r)$	$(2y + 1 + 30y^3)\left(-4 + y^3 + \frac{5}{4}y\right)$

2.7.6 Wrap-Up: Reflection

1. How "in the zone" do you feel right now about multiplying polynomials?

Unit 2, Lesson 8: Adding, Subtracting, and Multiplying Polynomials Using More Than One Operation

Benchmark

MA.912.AR.1.3 Add, subtract, and multiply polynomial expressions with rational number coefficients.

Learning Targets

- ☐ I can recognize adding, subtracting, or multiplying a polynomial will result in a polynomial.
- ☐ I can add, subtract, and multiply polynomial expressions using more than one operation.